1.1 Complex Numbers and Their Properties

INTRODUCTION No one person "invented" complex numbers, but controversies surrounding the use of these numbers existed in the sixteenth century. In their quest to solve polynomial equations by formulas involving radicals, early dabblers in mathematics were forced to admit that there were other kinds of numbers besides positive integers. Equations such as $x^2 + 2x + 2 = 0$ and $x^3 = 6x + 4$ that yielded "solutions" $1 + \sqrt{-1}$ and $\sqrt[3]{2 + \sqrt{-2}} + \sqrt[3]{2 - \sqrt{-2}}$ caused particular consternation within the community of fledgling mathematical scholars because *everyone* knew that there are no numbers such as $\sqrt{-1}$ and $\sqrt{-2}$, numbers whose square is negative. Such "numbers" exist only in one's imagination, or as one philosopher opined, "the imaginary, (the) bosom child of complex mysticism." Over time these "imaginary numbers" did not go away, mainly because mathematicians as a group are tenacious and some are even practical. A famous mathematician held that even though "they exist in our imagination ... nothing prevents us from ... employing them in calculations." Mathematicians also hate to throw anything away. After all, a memory still lingered that negative numbers at first were branded "fictitious." The concept of *number* evolved over centuries; gradually the set of numbers grew from just positive integers to include rational numbers, negative numbers, and irrational numbers. But in the eighteenth century the number concept took a gigantic evolutionary step forward when the German mathematician Carl Friedrich Gauss put the so-called imaginary numbers—or *complex numbers*, as they were now beginning to be called—on a logical and consistent footing by treating them as an extension of the real number system.

Our goal in this first section is to examine some basic definitions and the arithmetic of complex numbers.

The Imaginary Unit

Even after gaining wide respectability, through the seminal works of Carl Friedrich Gauss and the French mathematician Augustin Louis Cauchy, the unfortunate name "imaginary" has survived down the centuries. The symbol i was originally used as a disguise for the embarrassing symbol $\sqrt{-1}$. We now say that i is the **imaginary unit** and define it by the property $i^2 = -1$. Using the imaginary unit, we build a general complex number out of two real numbers.

DEFINITION 1.1.1 Complex Number

A **complex number** is any number of the form $z = a + ib$ where a and b are real numbers and i is the imaginary unit.

Terminology

The notations $a + ib$ and $a + bi$ are used interchangeably. The real number a in $z = a + ib$ is called the **real part** of z; the real number b is called the **imaginary part** of z. The real and imaginary parts of a complex number z are abbreviated $\text{Re}(z)$ and $\text{Im}(z)$, respectively. For example, if $z = 4 - 9i$, then $\text{Re}(z) = 4$ and $\text{Im}(z) = -9$. A real constant multiple of the imaginary unit is called a **pure imaginary number**. For example, $z = 6i$ is a pure imaginary number. Two complex numbers are **equal** if their corresponding real and imaginary parts are equal. Since this simple concept is sometimes useful, we formalize the last statement in the next definition.

Note: The imaginary part of $z = 4 - 9i$ is -9 **not** $-9i$. ▶

DEFINITION 1.1.2 Equality

Complex numbers $z_1 = a_1 + ib_1$ and $z_2 = a_2 + ib_2$ are **equal**, $z_1 = z_2$, if $a_1 = a_2$ and $b_1 = b_2$.

In terms of the symbols $\text{Re}(z)$ and $\text{Im}(z)$, Definition 1.1.2 states that $z_1 = z_2$ if $\text{Re}(z_1) = \text{Re}(z_2)$ and $\text{Im}(z_1) = \text{Im}(z_2)$.

The totality of complex numbers or the set of complex numbers is usually denoted by the symbol $\mathbf{C}$. Because *any* real number a can be written as $z = a + 0i$, we see that the set $\mathbf{R}$ of real numbers is a subset of $\mathbf{C}$.

Arithmetic Operations Complex numbers can be added, subtracted, multiplied, and divided. If $z_1 = a_1 + ib_1$ and $z_2 = a_2 + ib_2$, these operations are defined as follows.

Addition: $$z_1 + z_2 = (a_1 + ib_1) + (a_2 + ib_2) = (a_1 + a_2) + i(b_1 + b_2)$$

Subtraction: $$z_1 - z_2 = (a_1 + ib_1) - (a_2 + ib_2) = (a_1 - a_2) + i(b_1 - b_2)$$

Multiplication: $$\begin{aligned} z_1 \cdot z_2 &= (a_1 + ib_1)(a_2 + ib_2) \\ &= a_1a_2 - b_1b_2 + i(b_1a_2 + a_1b_2) \end{aligned}$$

Division: $$\begin{aligned} \frac{z_1}{z_2} &= \frac{a_1 + ib_1}{a_2 + ib_2},\ a_2 \neq 0, \text{ or } b_2 \neq 0 \\ &= \frac{a_1a_2 + b_1b_2}{a_2^2 + b_2^2} + i\frac{b_1a_2 - a_1b_2}{a_2^2 + b_2^2} \end{aligned}$$

The familiar commutative, associative, and distributive laws hold for complex numbers:

Commutative laws: $$\begin{cases} z_1 + z_2 = z_2 + z_1 \\ z_1z_2 = z_2z_1 \end{cases}$$

Associative laws: $$\begin{cases} z_1 + (z_2 + z_3) = (z_1 + z_2) + z_3 \\ z_1(z_2z_3) = (z_1z_2)z_3 \end{cases}$$

Distributive law: $$z_1(z_2 + z_3) = z_1z_2 + z_1z_3$$

In view of these laws, there is no need to memorize the definitions of addition, subtraction, and multiplication.

Addition, Subtraction, and Multiplication

(*i*) *To add (subtract) two complex numbers, simply add (subtract) the corresponding real and imaginary parts.*

(*ii*) *To multiply two complex numbers, use the distributive law and the fact that* $i^2 = -1$.

The definition of division deserves further elaboration, and so we will discuss that operation in more detail shortly.

EXAMPLE 1 Addition and Multiplication

If $z_1 = 2 + 4i$ and $z_2 = -3 + 8i$, find (**a**) $z_1 + z_2$ and (**b**) $z_1 z_2$.

Solution (**a**) By adding real and imaginary parts, the sum of the two complex numbers z_1 and z_2 is

$$z_1 + z_2 = (2 + 4i) + (-3 + 8i) = (2 - 3) + (4 + 8)i = -1 + 12i.$$

(**b**) By the distributive law and $i^2 = -1$, the product of z_1 and z_2 is

$$\begin{aligned} z_1 z_2 = (2 + 4i)(-3 + 8i) &= (2 + 4i)(-3) + (2 + 4i)(8i) \\ &= -6 - 12i + 16i + 32i^2 \\ &= (-6 - 32) + (16 - 12)i = -38 + 4i. \end{aligned}$$

Zero and Unity The **zero** in the complex number system is the number $0+0i$ and the **unity** is $1+0i$. The zero and unity are denoted by 0 and 1, respectively. The zero is the **additive identity** in the complex number system since, for any complex number $z = a + ib$, we have $z + 0 = z$. To see this, we use the definition of addition:

$$z + 0 = (a + ib) + (0 + 0i) = a + 0 + i(b + 0) = a + ib = z.$$

Similarly, the unity is the **multiplicative identity** of the system since, for any complex number z, we have

$$z \cdot 1 = (a + ib) \cdot (1 + 0i) = a - 0 + i(b - 0) = a + ib = z.$$

Conjugate If z is a complex number, the number obtained by changing the sign of its imaginary part is called the **complex conjugate**, or simply **conjugate**, of z and is denoted by the symbol $\bar{z}$. In other words, if $z = a + ib$, then its conjugate is $\bar{z} = a - ib$. For example, if $z = 6 + 3i$, then $\bar{z} = 6 - 3i$; if $z = -5 - i$, then $\bar{z} = -5 + i$. If z is a real number, say, $z = 7$, then $\bar{z} = 7$. From the definitions of addition and subtraction of complex numbers, it is readily shown that the conjugate of a sum and difference of two complex numbers is the sum and difference of the conjugates:

$$\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2, \quad \overline{z_1 - z_2} = \bar{z}_1 - \bar{z}_2. \tag{1}$$

Moreover, we have the following three additional properties:

$$\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2, \quad \overline{\left(\frac{z_1}{z_2}\right)} = \frac{\bar{z}_1}{\bar{z}_2}, \quad \bar{\bar{z}} = z. \tag{2}$$

Of course, the conjugate of any finite sum (product) of complex numbers is the sum (product) of the conjugates.

The definitions of addition and multiplication show that the sum and product of a complex number z with its conjugate $\bar{z}$ is a real number:

$$z + \bar{z} = (a + ib) + (a - ib) = 2a \tag{3}$$

$$z\bar{z} = (a + ib)(a - ib) = a^2 - i^2 b^2 = a^2 + b^2. \tag{4}$$

The difference of a complex number z with its conjugate $\bar{z}$ is a pure imaginary number:

$$z - \bar{z} = (a + ib) - (a - ib) = 2ib. \tag{5}$$

Since $a = \text{Re}(z)$ and $b = \text{Im}(z)$, (3) and (5) yield two useful formulas:

$$\text{Re}(z) = \frac{z + \bar{z}}{2} \quad \text{and} \quad \text{Im}(z) = \frac{z - \bar{z}}{2i}. \tag{6}$$

However, (4) is *the* important relationship in this discussion because it enables us to approach division in a practical manner.

Division

To divide z_1 by z_2, multiply the numerator and denominator of z_1/z_2 by the conjugate of z_2. That is,

$$\frac{z_1}{z_2} = \frac{z_1}{z_2} \cdot \frac{\bar{z}_2}{\bar{z}_2} = \frac{z_1\bar{z}_2}{z_2\bar{z}_2} \tag{7}$$

and then use the fact that $z_2\bar{z}_2$ is the sum of the squares of the real and imaginary parts of z_2.

The procedure described in (7) is illustrated in the next example.

EXAMPLE 2 Division

If $z_1 = 2 - 3i$ and $z_2 = 4 + 6i$, find z_1/z_2.

Solution We multiply numerator and denominator by the conjugate $\bar{z}_2 = 4 - 6i$ of the denominator $z_2 = 4 + 6i$ and then use (4):

$$\frac{z_1}{z_2} = \frac{2 - 3i}{4 + 6i} = \frac{2 - 3i}{4 + 6i}\frac{4 - 6i}{4 - 6i} = \frac{8 - 12i - 12i + 18i^2}{4^2 + 6^2} = \frac{-10 - 24i}{52}.$$

Because we want an answer in the form $a + bi$, we rewrite the last result by dividing the real and imaginary parts of the numerator $-10 - 24i$ by 52 and reducing to lowest terms:

$$\frac{z_1}{z_2} = -\frac{10}{52} - \frac{24}{52}i = -\frac{5}{26} - \frac{6}{13}i. \qquad \equiv$$

Inverses In the complex number system, every number z has a unique **additive inverse**. As in the real number system, the additive inverse of $z = a + ib$ is its *negative*, $-z$, where $-z = -a - ib$. For any complex number z, we have $z + (-z) = 0$. Similarly, every *nonzero* complex number z has a **multiplicative inverse**. In symbols, for $z \neq 0$ there exists one and only one nonzero complex number z^{-1} such that $zz^{-1} = 1$. The multiplicative inverse z^{-1} is the same as the **reciprocal** $1/z$.

EXAMPLE 3 Reciprocal

Find the reciprocal of $z = 2 - 3i$.

Solution By the definition of division we obtain

$$\frac{1}{z} = \frac{1}{2-3i} = \frac{1}{2-3i}\,\frac{2+3i}{2+3i} = \frac{2+3i}{4+9} = \frac{2+3i}{13}.$$

Answer should be in the form $a+ib$. ▶ That is,

$$\frac{1}{z} = z^{-1} = \frac{2}{13} + \frac{3}{13}i.$$

You should take a few seconds to verify the multiplication

$$zz^{-1} = (2-3i)\left(\tfrac{2}{13} + \tfrac{3}{13}i\right) = 1.$$

☰

Remarks COMPARISON WITH REAL ANALYSIS

(*i*) Many of the properties of the real number system **R** hold in the complex number system **C**, but there are some truly remarkable differences as well. For example, the concept of *order* in the real number system does not carry over to the complex number system. In other words, we cannot compare two complex numbers $z_1 = a_1 + ib_1$, $b_1 \neq 0$, and $z_2 = a_2 + ib_2$, $b_2 \neq 0$, by means of inequalities. Statements such as $z_1 < z_2$ or $z_2 \geq z_1$ have no meaning in **C** except in the special case when the two numbers z_1 and z_2 are real. See Problem 55 in Exercises 1.1. Therefore, if you see a statement such as $z_1 = \alpha z_2$, $\alpha > 0$, it is implicit from the use of the inequality $\alpha > 0$ that the symbol α represents a real number.

(*ii*) Some things that we take for granted as impossible in real analysis, such as $e^x = -2$ and $\sin x = 5$ when x is a real variable, are perfectly correct and ordinary in complex analysis when the symbol x is interpreted as a complex variable. See Example 3 in Section 4.1 and Example 2 in Section 4.3.

We will continue to point out other differences between real analysis and complex analysis throughout the remainder of the text.

EXERCISES 1.1

Answers to selected odd-numbered problems begin on page ANS-1.

1. Evaluate the following powers of i.

(a) i^8 **(b)** i^{11}

(c) i^{42} **(d)** i^{105}

2. Write the given number in the form $a+ib$.

(a) $2i^3 - 3i^2 + 5i$ **(b)** $3i^5 - i^4 + 7i^3 - 10i^2 - 9$

(c) $\dfrac{5}{i} + \dfrac{2}{i^3} - \dfrac{20}{i^{18}}$ **(d)** $2i^6 + \left(\dfrac{2}{-i}\right)^3 + 5i^{-5} - 12i$

In Problems 3–20, write the given number in the form $a + ib$.

3. $(5 - 9i) + (2 - 4i)$

4. $3(4 - i) - 3(5 + 2i)$

5. $i(5 + 7i)$

6. $i(4 - i) + 4i(1 + 2i)$

7. $(2 - 3i)(4 + i)$

8. $\left(\frac{1}{2} - \frac{1}{4}i\right)\left(\frac{2}{3} + \frac{5}{3}i\right)$

9. $3i + \dfrac{1}{2 - i}$

10. $\dfrac{i}{1 + i}$

11. $\dfrac{2 - 4i}{3 + 5i}$

12. $\dfrac{10 - 5i}{6 + 2i}$

13. $\dfrac{(3 - i)(2 + 3i)}{1 + i}$

14. $\dfrac{(1 + i)(1 - 2i)}{(2 + i)(4 - 3i)}$

15. $\dfrac{(5 - 4i) - (3 + 7i)}{(4 + 2i) + (2 - 3i)}$

16. $\dfrac{(4 + 5i) + 2i^3}{(2 + i)^2}$

17. $i(1 - i)(2 - i)(2 + 6i)$

18. $(1 + i)^2(1 - i)^3$

19. $(3 + 6i) + (4 - i)(3 + 5i) + \dfrac{1}{2 - i}$

20. $(2 + 3i)\left(\dfrac{2 - i}{1 + 2i}\right)^2$

In Problems 21–24, use the **binomial theorem***

$$(A + B)^n = A^n + \frac{n}{1!}A^{n-1}B + \frac{n(n-1)}{2!}A^{n-2}B^2 + \cdots + \frac{n(n-1)(n-2)\cdots(n-k+1)}{k!}A^{n-k}B^k + \cdots + B^n,$$

where $n = 1,\ 2,\ 3,\ \ldots$, to write the given number in the form $a + ib$.

21. $(2 + 3i)^2$

22. $\left(1 - \frac{1}{2}i\right)^3$

23. $(-2 + 2i)^5$

24. $(1 + i)^8$

In Problems 25 and 26, find $\mathrm{Re}(z)$ and $\mathrm{Im}(z)$.

25. $z = \left(\dfrac{i}{3 - i}\right)\left(\dfrac{1}{2 + 3i}\right)$

26. $z = \dfrac{1}{(1 + i)(1 - 2i)(1 + 3i)}$

In Problems 27–30, let $z = x + iy$. Express the given quantity in terms of x and y.

27. $\mathrm{Re}(1/z)$

28. $\mathrm{Re}(z^2)$

29. $\mathrm{Im}(2z + 4\bar{z} - 4i)$

30. $\mathrm{Im}(\bar{z}^2 + z^2)$

In Problems 31–34, let $z = x + iy$. Express the given quantity in terms of the symbols $\mathrm{Re}(z)$ and $\mathrm{Im}(z)$.

31. $\mathrm{Re}(iz)$

32. $\mathrm{Im}(iz)$

33. $\mathrm{Im}((1 + i)z)$

34. $\mathrm{Re}(z^2)$

In Problems 35 and 36, show that the indicated numbers satisfy the given equation. In each case explain why additional solutions can be found.

35. $z^2 + i = 0$, $z_1 = -\dfrac{\sqrt{2}}{2} + \dfrac{\sqrt{2}}{2}i$. Find an additional solution, z_2.

36. $z^4 = -4$; $z_1 = 1 + i,\ z_2 = -1 + i$. Find two additional solutions, z_3 and z_4.

---

*Recall that the coefficients in the expansions of $(A + B)^2$, $(A + B)^3$, and so on, can also be obtained using Pascal's triangle.

In Problems 37–42, use Definition 1.1.2 to solve each equation for $z = a + ib$.

37. $2z = i(2 + 9i)$

38. $z - 2\bar{z} + 7 - 6i = 0$

39. $z^2 = i$

40. $\bar{z}^2 = 4z$

41. $z + 2\bar{z} = \dfrac{2 - i}{1 + 3i}$

42. $\dfrac{z}{1 + \bar{z}} = 3 + 4i$

In Problems 43 and 44, solve the given system of equations for z_1 and z_2.

43.
$$\begin{aligned} iz_1 - \quad & iz_2 = 2 + 10i \\ -z_1 + (1 - i)z_2 &= 3 - 5i \end{aligned}$$

44.
$$\begin{aligned} iz_1 + (1 + i)z_2 &= 1 + 2i \\ (2 - i)z_1 + \quad & 2iz_2 = 4i \end{aligned}$$

Focus on Concepts

45. What can be said about the complex number z if $z = \bar{z}$? If $(z)^2 = (\bar{z})^2$?

46. Think of an alternative solution to Problem 24. Then without doing any significant work, evaluate $(1 + i)^{5404}$.

47. For n a nonnegative integer, i^n can be one of four values: 1, i, -1, and $-i$. In each of the following four cases, express the integer exponent n in terms of the symbol k, where $k = 0, 1, 2, \ldots$.

(**a**) $i^n = 1$

(**b**) $i^n = i$

(**c**) $i^n = -1$

(**d**) $i^n = -i$

48. There is an alternative to the procedure given in (7). For example, the quotient $(5 + 6i)/(1 + i)$ must be expressible in the form $a + ib$:

$$\frac{5 + 6i}{1 + i} = a + ib.$$

Therefore, $5 + 6i = (1 + i)(a + ib)$. Use this last result and Definition 1.1.2 to find the given quotient. Use this method to find the reciprocal of $3 - 4i$.

49. Assume for the moment that $\sqrt{1 + i}$ makes sense in the complex number system. How would you then demonstrate the validity of the equality

$$\sqrt{1 + i} = \sqrt{\tfrac{1}{2} + \tfrac{1}{2}\sqrt{2}} + i\sqrt{-\tfrac{1}{2} + \tfrac{1}{2}\sqrt{2}}?$$

50. Suppose that z_1 and z_2 are complex numbers. Use Definition 1.1.2 to prove that if $z_1 z_2 = 0$, then $z_1 = 0$ or $z_2 = 0$.

51. Suppose the product $z_1 z_2$ of two complex numbers is a nonzero real constant. Show that $z_2 = k\bar{z}_1$, where k is a real number.

52. Without doing any significant work, explain why it follows immediately from (2) and (3) that $z_1\bar{z}_2 + \bar{z}_1 z_2 = 2\text{Re}(z_1\bar{z}_2)$.

53. Mathematicians like to prove that certain "things" within a mathematical system are unique. For example, a proof of a proposition such as "The unity in the complex number system is unique" usually starts out with the assumption that there exist two *different* unities, say, 1_1 and 1_2, and then proceeds to show that this assumption leads to some contradiction. Give one contradiction if it is assumed that two different unities exist.

54. Follow the procedure outlined in Problem 53 to prove the proposition "The zero in the complex number system is unique."

55. A number system is said to be an **ordered system** provided it contains a subset P with the following two properties:

First, for any nonzero number x in the system, either x or $-x$ is (but not both) in P.